

Solar Panel Behavior

Valparaiso University - Electrical and Computer Engineering

Solar energy from the sun is about a 1000 W/m^2 at the top of the Earth's atmosphere.
How can we capture that energy?

1. Lamp info

Lamp name / type _____ incandescent | LED

Rated input power _____ W

Rated brightness _____ lm

Measured input power _____ W

2. Solar panel info

Num. cells in series _____

Open circuit voltage _____ V

→ *computed* _____ V / cell

3. Shading cells

The series connection of the individual solar cells into the complete panel has consequences for how partial shade affects the panel's output.

Measure short-circuit current _____ mA

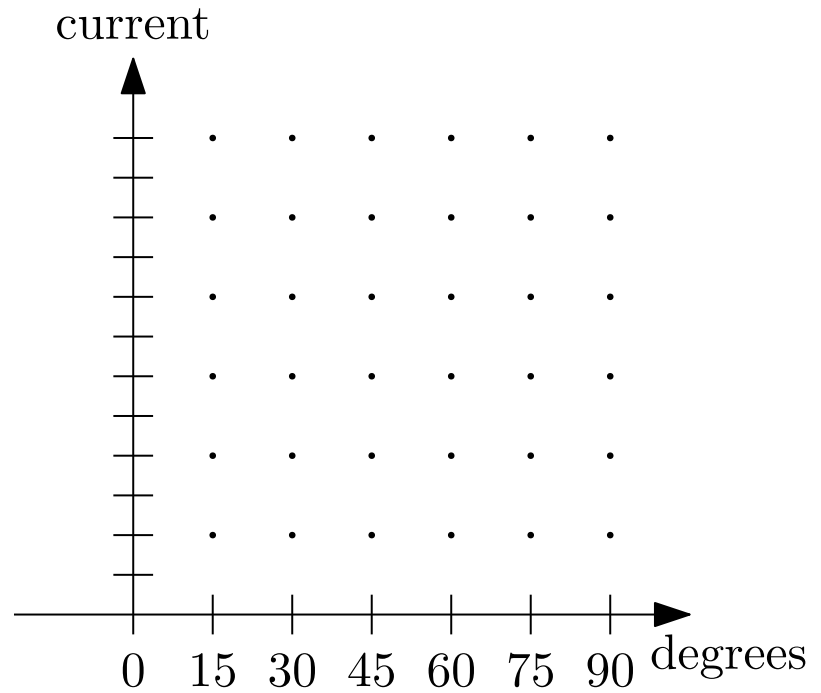
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- ☐ Block panel slowly from one direction while watching the measured current.

 - ☐ Block panel slowly from 90° to the last direction, watching the measured current.

4. Current vs. Angle

The short-circuit current represents the power generated by a solar panel (multiplied by the “loaded” voltage gives power in watts). The optimum voltage decreases a bit when the temperature of the solar cell increases.

- ☐ Mark the measured short-circuit current value for 0° tilt angle and **decide your y-axis scale marks.**
- ☐ Tilt the panel up to about 15° and mark on the plot the new short-circuit current. Estimate the angle by moving it to 45° and then splitting into thirds.
- ☐ Continue to tilt the panel to each angle and record the current.





DO NOT PRINT this and following pages.
Only print the first two pages for the student handout.

This page is found at agnd.net/valpo/solar/panel-lamp and agnd.net/valpo/solar/panel-lamp.pdf

5. Logistics

5.1. Support hardware NOT in the box

- ☐ Banana to mini-grabber leads. 2× per station.
- ☐ Multimeter per station. (bannana-grabbers are the probes)
- ☐ Printed worksheet handout.
- ☐ Pencils / pens for **long** version of the activity.

5.2. Configuration

6 stations. Pairs of comparisons with one difference.

1. LED panel-lamp with large blue solar panel
2. Incandescent with large blue solar panel
3. LED "300W" with large blue solar panel
4. Incandescent with medium blue solar panel
5. Incandescent with small red solar panel
6. Incandescent with small red solar panel

5.3. Supplies in the Box

Listed in order of packing the box.

- ☐ Extra printed handouts
- ☐ 3× large blue solar panels
- ☐ 1× medium blue solar panels
- ☐ 2× small red solar panels
- ☐ 6× USB to terminal breakout cables
- ☐ 4× 3-ft extension cords
- ☐ 4× AC power meters
- ☐ 5× lamp reflectors - 300W Brooder Clamp Work Light . The bulbs need to be removed and put in the small box for storage so the reflectors can nest together and to reduce the likelihood of breaking a bulb.
- ☐ 3-panel LED garage lamp. ≈6000 lumen
- ☐ 1× LED "300W" bulb. 4060 lumen
- ☐ 4× incandescent bulbs. 200W, frosted or clear, 3650 or 3880 lumen
- ☐ Product boxes for the lamps showing power use and light output.

5.4. Operational notes

Pre-setup the multimeters to use a fixed range instead of auto-ranging. *Which* range depends on the particular lamp+panel combination.

There is *always* one multimeter with a blown fuse for the ammeter input.

Working through the entire worksheet can only be completed in 20min IFF students are already familiar with multimeter measurements and making + changing connections.

Typical issues:

- 60% don't immediately know how to use a mini-grabber to make the connection to the USB breakout wires. Give a demo of how they work. Better: include a pic in the handout procedure.
- Need student eye contact when showing how to **see** and **count** the number of cells in the panel. The flat silvered wires between cells **visually** look like they are splitting the cells themselves, so most people count the dark patches instead of the full-width cell.

5.5. HS demo activity

For: 20min max activity with high schoolers

Few are very good or quick at making connections to test equipment — minimize them changing the setup mid-activity.

5.5.1. Agenda

- Solar **panel** is made from solar **cells** plus support structure and wiring.
 - Each solar cell generates about 0.6 V or 600 mV, regardless of the individual cell's size/surface area.
 - Show the 250 W panel and point out an individual cell there.
 - With $12 \times 8 = 96$ cells in **series**, this is about 57.6 V expected. The label says 59.8 V, so really close (< 4% error)!
 - Show how to **identify** an individual cell on their panel. The wiring is flat and weaves over/under; look for the edges of the thin dark plate and don't be fooled by the wiring to over-count the number.
 - How many cells in your panel? (9 for the USB ones)
 - Compute the panel's expected voltage. $9 \times 0.6 \text{ V} = 5.4 \text{ V}$.
 - (skip to save time) Measure the open-circuit panel voltage with the DMM and compare to predicted.
 - The **exact voltage** drops when temperature **increases**, so a hot cell/panel measures less voltage.
 - The specific manufacturing of the cell will also change the O-C voltage a little.
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- Switch to measure **short-circuit current**. Save time by pre-staging this setup.
 - Put panel flat on table directly under the lamp. Move panel around a little to get the biggest current value. Red SunStream panel is about 100 mA with the 200 W incandescent bulb.
 - Take the thick paper and slowly cover **ONE** cell. See how the current drastically drops by 80%. Series connection and we closed one of the “valves”.
 - Slowly shade the panel from the right angle direction, covering parts of ALL the cells. The current smoothly decreases roughly proportional to the amount of shading.

This shading has consequences on how and where to mount solar panels. While it's possible to add extra electronic parts to bypass or detour current around a shaded cell, this costs money and hardware, and thus is often not done.

Can not have shading over any part of a panel. Similarly if the panels are series-connected. Sometimes the bypass diodes are only per panel instead of per cell.

Optional, measure the S-C current versus panel tilt angle.

What does this mean for you, the engineer, designing a *complete* system?

From above, it's "**best**" to ensure that your panel always directly faces the sun over the course of a day. But this requires some structures, bearings, and motors to move them in the right directions.

- Increased time to design the whole solar farm.
- More cost of the parts.
- More time to construct the farm.
- More labor for the construction.

Once installed, the mechanisms need to be maintained: grease the bearings, fix things that break, etc. Mowing the grass under the panels may be needed—remember shading the cells.

This is why most solar farms you see are just set at one angle and do not move.

What if you made the frames tall enough so they were taller than the weeds? Then: *no need to mow at all!* Balancing against this is that the frames now require more steel, which increases the install cost.

Engineering designers don't always choose the **most efficient** or "best" technical solution. We take into account the **entire** project, from design time and cost, to construction, to maintenance over the system's operational life. Money and time are always part of our "calculations".

What is **best** depends LOTS on the specific details of the situation and the needs or reasons for the project in the first place.